Efficient Large Language Model Deployment on Edge Devices

CSD415 Project Phase I Presentation

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Overview

Introduction

- Large Language Models (LLMs) such as GPT-4, PaLM, and LLaMA have revolutionized text understanding and generation.
- However, their massive scale (billions of parameters) prevents deployment on low-resource environments.
- Edge devices suffer from memory, compute, and power constraints.
- Bridging this gap requires compression pipelines that reduce model size while retaining linguistic and reasoning capabilities.

Problem Statement

- Cloud-based inference for LLMs incurs latency, privacy risks, and dependency on network availability.
- Running uncompressed models on-device leads to high inference delay and thermal throttling.
- Current compression methods often sacrifice accuracy or fail to generalize to multilingual and reasoning tasks.
- **Goal:** Enable practical and privacy-friendly on-device deployment of LLMs through an optimized hybrid compression framework.

Proposed Solution

- Build upon existing works in Knowledge Distillation (Acharya et al., Cantini et al.), Quantization (Lang et al.), and Pruning (Yi et al.).
- Develop a **hybrid compression pipeline** that:
  1. Distills semantic knowledge from large teacher models into smaller students.
  2. Applies structured pruning for attention heads and redundant layers.
  3. Employs mixed-precision quantization (8/4-bit) optimized for ARM and NPU hardware.
- Incorporate **Multi-Token Prediction (Samragh et al.)** to improve inference throughput.
- The approach aims to preserve up to 95% accuracy with an 80% reduction in model size.

Literature Survey Overview

- Extensive review conducted on 15+ recent works (2021–2025).
- Focused on pruning, quantization, and knowledge distillation (KD) techniques.
- Recent studies emphasize hybrid frameworks combining these methods for better trade-offs.
- Key trends:
    - **Symbolic Explainable KD** – improved interpretability.
    - **Hybrid Compression Pipelines** – joint pruning + quantization.
    - **Multi-token Inference** – improved throughput for LLMs.

Literature Survey Summary (2023–2025)

No.	Year	Name of Paper	Methodology	Advantage	Disadvantage
1	2024	Symbolic KD for LLMs (Acharya et al.)	Interpretable targets	Better interpretability	High complexity
2	2024	XAI-Driven KD for Low-Resource Devices (Cantini et al.)	Explainable KD	Transparent	Dependent on XAI quality
3	2023	Quantization Robust Pruning (Kim)	Pruning + KD	High compression	Complex tuning
4	2023	Mutual Learning Framework (JCST Team)	Multi-network KD	Stable training	Resource overhead
5	2023	Efficient Hybrid Compression (Malihi et al.)	KD + Pruning + Fine-Tuning	Accuracy control	Depends on fine-tuning
6	2023	Quantization Techniques for LLMs (Lang et al.)	Comparative study	Hardware insights	Accuracy drop risk
7	2023	Advances in Pruning and Quantization (Bibi et al.)	Survey	Broad coverage	Limited LLM data

Table: Key Works in LLM Compression (Part I)

Literature Survey Summary (2023–2025 Continued)

No.	Year	Name of Paper	Methodology	Advantage	Disadvantage
8	2023	Linear Quantization Optimization (Algorithms Team)	Low-bit quantization	Efficient edge use	Hardware dependency
9	2023	Federated Layer-wise Pruning (Entropy Team)	Layer pruning in FL	Adaptive edge design	Comm. trade-offs
10	2025	EdgeMoE (Yi et al.)	Sparse mixture-of-experts	High sparsity	Loss of expressiveness
11	2024	Adaptive Scaling Pruning (Ko, Kang)	Filter pruning for vision	Efficient edge vision	Domain-specific
12	2024	Hybrid Weighted Pruning (Geng et al.)	Weighted hybrid pruning	High FLOP reduction	Fine-tuning needed
13	2023	Statistical Pruning for IoT (Mao et al.)	Statistical pruning	Low power usage	IoT-specific
14	2021	Sparsity in Deep Learning (Hoefler et al.)	Growth-pruning survey	Contextual pruning insight	No LLM focus
15	2025	Multi-Token Prediction (Samragh et al.)	Multi-token decoding	Faster inference	Experimental

Table: Key Works in LLM Compression (Part II)

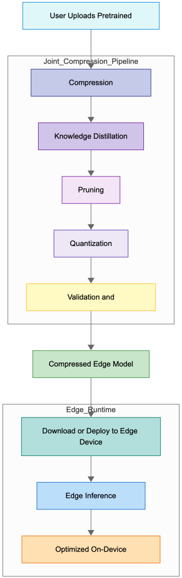
Objectives

- Design a **modular compression framework** integrating Knowledge Distillation (KD), pruning, and quantization.
- Evaluate **compression-accuracy trade-offs** for different LLM architectures.
- Develop an **adaptive runtime** for efficient inference across diverse edge hardware.
- Establish **reproducible pipelines** with defined benchmarks for latency, throughput, and accuracy.

Expected Outcomes

- **Efficiency:** 80% model compression, 2–3$\times$ faster inference.
- **Scalability:** Portable across CPU, GPU, and NPU.
- **Interpretability:** Distilled models retain explainable attention patterns.
- **Edge-readiness:** Suitable for on-device AI assistants and IoT endpoints.

System Architecture



Software Requirements Specification

- **Functional Requirements:**
  - Support KD, pruning, and quantization modules.
  - APIs for compression, deployment, and profiling.
- **Non-Functional Requirements:**
  - Latency ¡ 150 ms (first token)
  - Throughput 20 tokens/s
  - Memory footprint ¡ 1.5 GB
- Design documents, SRS, and architecture blueprints have been prepared to formalize implementation.

System Design Overview (Part I)

- Combines symbolic KD (Acharya et al.) with quantization-aware training (Lang et al.).
- Supports structured pruning guided by model saliency maps.
- Utilizes a flexible recipe-based configuration for hardware adaptation.
- Edge runtime supports CPU, GPU, and NPU backends with dynamic precision scaling.

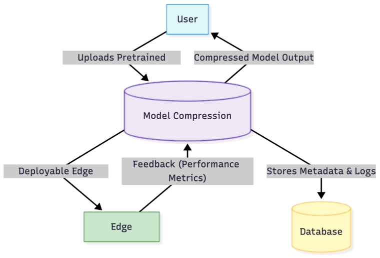


Figure: Overall System Architecture

System Design Overview (Part II)

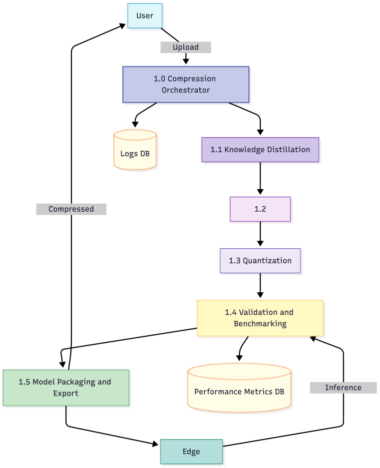


Figure: Edge Runtime Flow Diagram

Conclusion

- The study focused on enabling efficient LLM deployment on resource-constrained environments.
- Existing literature demonstrated strong potential in hybrid compression methods.
- Building on these insights, our project designs a practical pipeline combining KD, pruning, and quantization.
- We have completed SRS, design documentation, and architectural planning for the proposed model.
- Future phases will focus on implementation, benchmarking, and real device testing.

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Thank You!